

Elastic wave propagation in a half space

Chapter 5

The previous chapter explained Green's function in an infinite medium. However, the structure of the real Earth is not so simple. Lateral heterogeneity causes the complexity of seismic wave propagation. This figure shows SH wave propagation from an earthquake. At frequencies below a few Hz, a stratified Earth structure (a seismic wave velocity structure depending only on depth) is a good approximation. Even the simple Earth structure still exhibits complexities, but we can trace the wavefront. The figure shows the waves can be approximated by plane waves.

In this figure, we can see reflections and conversions on the discontinuities (at the surface, 410 km, 660 km, and core-mantle boundary (CMB)). Because the free surface is the biggest boundary, we introduce the free surface first in this chapter.

First, we introduce the concept of a plane wave. Then, we will show that we can separate the wavefield into P-SV and SH waves according to the wave type and the polarization direction. This chapter describes the effects of a free surface: the reflection and the P-S conversion at a free surface.

§5.1 Review of seismic wave propagation: body waves and surface and boundary waves

Nature is full of waves. For example, if we look at the water's surface, we can see ripples spreading as the wind blows. When some restoring force acts on the medium, they propagate at a certain speed while maintaining their shape ($f(x - ct)$ (x is position, c is propagation speed, t is time)). Let us consider sound waves specifically.

1. Gas moves and density changes
2. Density change produces pressure change
3. A pressure gradient moves the gas

Sound waves propagate by repeating this cycle. Seismic waves propagate in solids in a similar cycle (see the next chapter for details). If the wave's amplitude is sufficiently small, its behavior is linear, and the principle of superposition holds. In other words, if individual wave components can be isolated and understood, the entire wavefield can be understood as a superposition of these components.

§5.2 Plane wave

To be revised Elastic wave propagation becomes more complex when considering free surfaces compared to a homogeneous infinite medium. For example, when trying to evaluate the Green's function for a half-space with a free surface, it can only be expressed analytically in limited cases, such as when the source is located on the ground surface (e.g., chapter 5.8). Even in the case of the Green's function for a homogeneous infinite elastic medium, the analytical form of the near-field term becomes complex. When considering Green's function for the infinite medium, the near-field term changes shape depending on the distance from the epicenter, while the far-field term maintains its shape. On the other hand, the far-field term propagates while keeping its shape^{note 1)}. Therefore, this chapter will focus only on the far-field term.

Let us first consider the 3-dimensional scalar case for simplicity. Consider the case of an external force $\delta(\mathbf{x})$ acting at the epicenter. As we learned previously, the Green's function, in this case, is given^{note 2)} by

$$G^\phi(r, \omega) = -\frac{1}{4\pi\kappa} \frac{e^{-ikr}}{r}. \quad (5.1)$$

We can define an isosurface where the phase of e^{-ikr} is constant and is generally referred to as a "wavefront". The trajectory orthogonal to the wavefront is called the ray (see Chapter ?? for details). In regions where r is sufficiently distant, the wavefront curvature can be neglected.^{note 3)} The wave can be treated as a plane wave when $r > 1/k$, where r is the radius of curvature.

^{note 1)}As explained in the section on Ray Theory, the far-field term maintains its shape when the typical spatial scale of the velocity structure is longer than the wavelength of interest.

^{note 2)}In the 2-dimensional case as well as in the 3-dimensional case, the far-field of the Green's function can be approximated as e^{-ikr} 3.130.

^{note 3)}More exact conditions are required for the approximation of plane waves: (i) the curvature of the wavefront can be neglected when focusing on wave propagation on spatial scales sufficiently short compared to the radius of curvature of the wavefront. (ii) Near the epicenter, the radius of curvature becomes so small that a plane wave cannot approximate it. From a simple estimation, it can be seen that the wave can be treated as a plane wave in the region of approximately $r > 1/k$

Let us go back to the wave equation for once: the Green's function G^ϕ in the frequency domain satisfies

$$-\rho_0 \omega^2 G^\phi - \kappa \nabla^2 G^\phi = -\delta(\mathbf{x}). \quad (5.2)$$

The Fourier transform in the space, and Fourier inverse transform again gives

$$G^\phi = -\frac{1}{\kappa} \iiint \frac{\alpha^2 e^{i(k_x x + k_y y + k_z z)}}{\alpha^2(k_x^2 + k_y^2 + k_z^2) - \omega^2} dk_x dk_y dk_z \quad (5.3)$$

We can write that this equation shows that the Green's function can be represented by a superposition of plane waves.

When considering the Green's function of a semi-infinite medium, if we understand the behavior of the plane wave element $e^{-ik \cdot \mathbf{x}}$ at the free boundary surface, we can understand the overall behavior by superposition (inverse Fourier transform in the frequency domain). In other words, understanding the behavior of the plane wave $e^{-ik \cdot \mathbf{x}}$ on the free surface is the key to understanding elastic wave propagation in a semi-infinite medium.

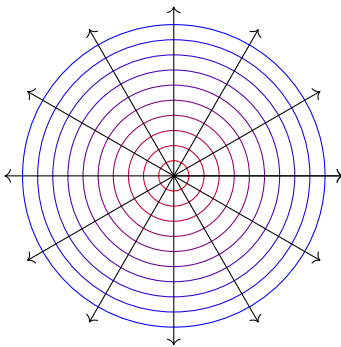


Figure 5.1: Wavefront and ray for Green's function for a 3-D in an infinite medium.

Summary of plane wave

When the source is enough far away from the source, the curvature of the wavefront becomes small. In this case, the waveform can be approximated by a plane wave:

1. Plane wave is preserving its shape,
2. Plane wave has a plane wavefront,
3. Plane wave propagates perpendicular to the wavefront^a.

In this case, the scalar variable $\phi(t - \mathbf{p} \cdot \mathbf{x})$ can be represented as a function of $t - \mathbf{k} \cdot \mathbf{x}$. $\mathbf{p}$ is a quantity called slowness defined as $\mathbf{p} = \mathbf{k}/\alpha$ (acoustic wave).

^aExactly speaking, here we neglect the dispersion for simplicity.

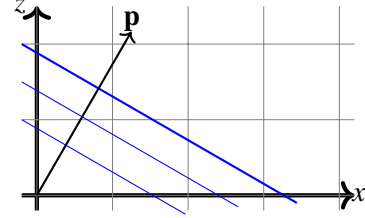
Here we consider a propagating wave into the x axis ($p_y = 0$). When $p_x > 1/\alpha$, p_z has real value. Then it propagates into the z direction (plane wave). with P-wave speed α along the slowness vector. On the ground ($z = 0$), at $t = 0$ the wavefront is at the point of $x = z = 0$. At $t = \tau$ it is at the point of $(\tau/p_x, 0, \tau/p_z)$. When we observed the waveform on the ground using seismometers, it propagates in the x direction with "apparent velocity" with $1/p_x$. Thus we can estimate the horizontal apparent velocity from the surface seismic observations.

5.2.1 Plane wave in an elastic medium

Next, Let us consider an elastic wave with a sinusoidal shape at angular frequency ω ^{note 4)}, and slowness vector $\mathbf{p}$. Scalar potential ϕ and vector potential ψ can be written by^{note 5)}

$$\phi = A_\alpha e^{i(\mathbf{k}_\alpha \cdot \mathbf{x} - \omega t)} = A_\alpha e^{i\omega(\mathbf{p}_\alpha \cdot \mathbf{x} - t)} \quad (5.4)$$

$$\psi = A_\beta e^{i(\mathbf{k}_\beta \cdot \mathbf{x} - \omega t)} = A_\beta e^{i\omega(\mathbf{p}_\beta \cdot \mathbf{x} - t)}. \quad (5.5)$$



The corresponding displacement is given by,

$$s_\alpha(\mathbf{x}, t) = A_\alpha \mathbf{k}_\alpha e^{i(\mathbf{k}_\alpha \cdot \mathbf{x} - \omega t)} = \omega A_\alpha \mathbf{p}_\alpha e^{i\omega(\mathbf{p}_\alpha \cdot \mathbf{x} - t)} \quad (5.6)$$

$$s_\beta(\mathbf{x}, t) = \mathbf{k}_\beta \times A_\beta e^{i(\mathbf{k}_\beta \cdot \mathbf{x} - \omega t)} = \mathbf{p}_\beta \times A_\beta \omega e^{i\omega(\mathbf{p}_\beta \cdot \mathbf{x} - t)}. \quad (5.7)$$

The polarization vector of the P wave is parallel to the propagation direction $\mathbf{p}_\alpha$, whereas that of the S wave is perpendicular to the propagation direction, with an ambiguity in its orientation. Although we use the form $e^{i\mathbf{k} \cdot \mathbf{x} - \omega t}$ here to represent wave propagation, we note that only the real part of a physical quantity has physical meaning.

$\mathbf{p}$ is called **slowness**, defined as $\mathbf{p} = \mathbf{k}/\omega$. Its dimension is the inverse of speed, and it is parallel to the propagation direction. For example, the squared norm can be written as

$$p_x^2 + p_y^2 + p_z^2 = \frac{1}{\alpha^2}. \quad (5.8)$$

Inverse Fourier transform of $e^{i\omega(\mathbf{p} \cdot \mathbf{x} - t)}$ gives us formula of general waveform. A P-wave propagates while maintaining its shape as $s_\alpha(\mathbf{x}, t) = \mathbf{p}_\alpha f(t - \mathbf{p}_\alpha \cdot \mathbf{x})$, whereas an S-wave propagates as $s_\beta(\mathbf{x}, t) = \mathbf{p}_\beta \times A_\beta f(t - \mathbf{p}_\beta \cdot \mathbf{x})$.

5.2.2 Body wave and inhomogeneous wave

^{note 4)}When we consider seismic wave propagation, the sign of the Fourier convention is different from other physical cases, including the appendix of the previous chapter. To keep consistency a propagation in positive x direction as $e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$. Read the box of Aki and Richards for details. Please take care of the Fourier convention when you read a paper of a textbook.

^{note 5)}In seismology, when considering wave propagation, the sign on ω is often taken to be negative. The definition of the Fourier transform is also often changed. This is to treat traveling waves as positive, and this is the sign taken by Aki and Richards (2002) and Saito (2016). In seismology, too, Dahlen and Tromp (1998) use the opposite sign, so it is important to be careful about which definition you are following.

When $p^2 > 1/\alpha^2$, p_z becomes imaginary. Using an definition of $p_z = \xi i$, the corresponding potential is given by

$$\phi = A_\alpha e^{i\omega(p_x x - \omega t)} e^{-\xi z}. \quad (5.9)$$

This equation shows that it decreases exponentially in the z direction. The wave is called an inhomogeneous wave^{note 6)} In an infinite medium, it diverges at infinity, making it physically impossible as a standalone wave. However, when boundaries exist, it can exist while satisfying the boundary conditions. Section 5.6 explains Rayleigh waves as a kind of inhomogeneous wave in a semi-infinite medium with a free boundary.

Let us review the global propagation of seismic waves. We can categorize it into body wave, which propagates in the Earth's interior, and boundary/surface wave, which propagates along a boundary including the Earth's surface (Figure 5.2left). Figure 5.2 shows such an example of Green's functions using seismic interferometry. The Horizontal axis shows the epicentral distance, whereas the vertical one shows the travel time^{note 7)}. The figures show global propagations of seismic surface waves and body waves. This section describes a brief summary of seismic wave types.

In general, we can categorize seismic waves into body waves and boundary waves. Body waves propagate in an internal body, whereas boundary waves travel along a boundary. A surface wave is a kind of boundary wave trapped close to the free surface.

^{note 6)} This wave is also known as an evanescent wave or external wave in physics or meteorology.

^{note 7)} travel time is defined as the time from the origin time to the arrival time

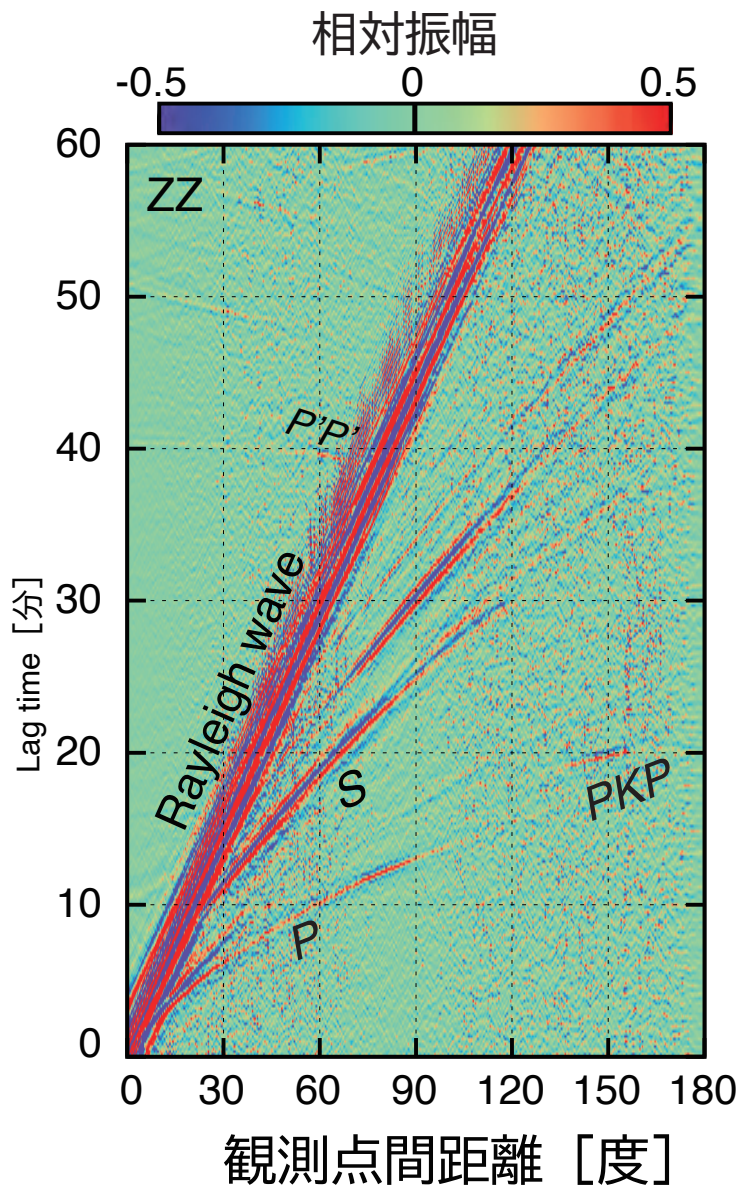


Figure 5.2: Global propagation of body and surface waves. The waveforms are virtual Green's functions retrieved by cross-correlating ambient seismic wave field, also known as microseisms⁽³⁾ (see chapter 10 for details of seismic interferometry.)

Body wave

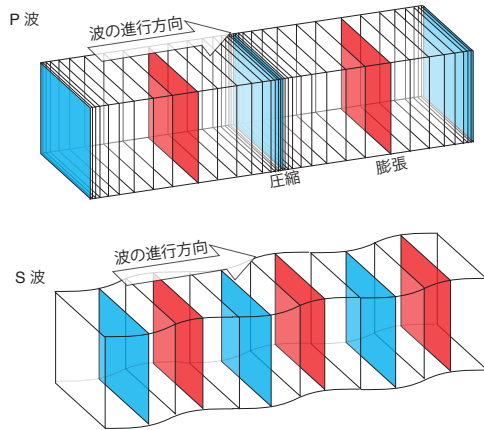


Figure 5.3: A schematic figure of P and S wave propagation.

We can categorize body waves into P and S waves; the polarization of the P wave is parallel to the propagation direction, whereas that of the S wave is perpendicular to the propagation direction (Figure 5.3). "P" originated from the Primary wave, and "S" originated from the Secondary wave. The deformation of the P wave is volumetric, whereas that of the S wave is shear. ^{note 8)} In general, with the decreasing temperature of the material, the stiffness increases, which causes an increase in the P wave and S wave velocities.

We all know well that P-waves propagate faster than S-waves, and the Omori formula for determining the distance to the epicenter from the difference in the arrival times of P-waves and S-waves ^{note 9)}. You may have also heard of the Earthquake Early Warning, which estimates the hypocenter from the fast-moving P-waves and predicts the arrival of

large tremors (S-waves).

Inhomogeneous waves: surface waves and boundary wave

Surface waves in an elastic medium can be categorized into Rayleigh waves, associated with volumetric changes, and Love waves associated with multiple reflections of SH waves in a surface low-velocity layer.

5.2.3 Energy flux

The total energy of an elastic medium per unit volume $\mathcal{U}$ can be written as the sum of the kinetic energy $\mathcal{U}^k$ and the strain energy $\mathcal{U}^p$ as

$$\mathcal{U} = \mathcal{U}^k + \mathcal{U}^p. \quad (5.10)$$

^{note 8)} A Web site of demonstration of body wave propagation http://www.eri.u-tokyo.ac.jp/knishida/Seismology/body_wave.html

^{note 9)} ⁽⁴⁾ is available at <http://hdl.handle.net/2261/32677>

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

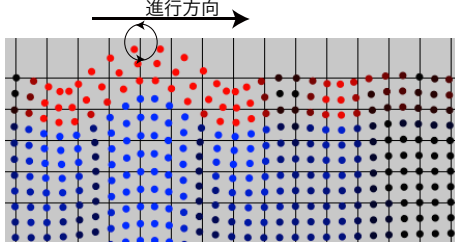


Figure 5.4: Rayleigh wave propagation. The red color shows particle motions in retrograde, and the blue color shows those in prograde.

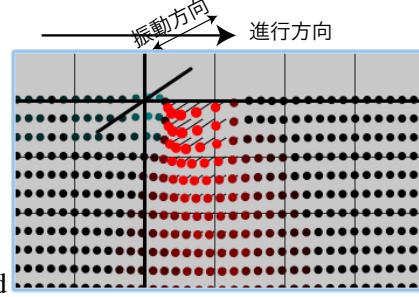


Figure 5.5: Love wave propagation.

Energy flux $\mathbf{K}$ perpendicular to a unit area is given by⁽¹⁾

$$\mathbf{K} = -\mathbf{T} \cdot \partial_t \mathbf{s}. \quad (5.11)$$

By equipartition between the kinetic energy and the strain energy, the total energy $\mathcal{U}$ can be written by,

$$\mathcal{U} = \rho \left| \frac{\partial s}{\partial t} \right| \quad (5.12)$$

where A is the displacement.

In particular, the energy flux of the P wave and that of the S wave are given by

$$\mathbf{K} = \begin{cases} \alpha \hat{\mathbf{n}} \mathcal{U} & \text{P wave} \\ \beta \hat{\mathbf{n}} \mathcal{U} & \text{S wave,} \end{cases} \quad (5.13)$$

where $\hat{\mathbf{n}}$ is a unit vector of the propagation direction. Conservation of energy is represented by

$$\frac{\partial \mathcal{U}}{\partial t} + \nabla \cdot \mathbf{K} = 0. \quad (5.14)$$

Problem 5.1

1. For a plane wave (P wave), show that the kinetic energy is equal to the strain energy as,

$$\frac{1}{2}T_{ij}E_{ij} = \frac{1}{2}\rho \left| \frac{\partial \mathbf{s}}{\partial t} \right|^2. \quad (5.15)$$

2. Here we consider a wave form of P wave $\mathbf{s} = \mathbf{p} f(t - \mathbf{p} \cdot \mathbf{x})$. Calculate the energy flux from equation 5.11.
3. For the inhomogeneous wave (P wave) with propagation in $z < 0$ given by P wave potential $\phi = \sin(\omega t - kx) \exp(\xi z)$. Then the displacement is written by

$$s_x = -k \cos(\omega t - kx) \exp(\xi z) \quad (5.16)$$

$$s_z = \xi \sin(\omega t - kx) \exp(\xi z). \quad (5.17)$$

Calculate the energy flux.

§5.3 SH wave and P-SV wave

Free surface condition on the ground is crucial for seismic wave propagations. For simplicity, let us consider semi-infinite homogeneous medium in $z < 0$ with free surface condition on $z = 0$ ($T_{iz}=0$) (Figure 5.7). By introducing the free surface, we can categorize S wave into two. Figure 5.6 shows (1) vertically polarized S wave in xz plane (SV wave) and (2) horizontally polarized S wave (SH wave). This category is also crucial for stratified Earth, which is a good approximation at frequencies lower than several Hz.

In this chapter, the first section explains reflections and refraction of SH wave, and then we will explain reflection, refraction, and conversion of SV waves.

5.3.1 Equations of motion and Hooke's law

As shown in Figure 5.6, we take the y axis along the wave front,

$$\frac{\partial}{\partial y} = 0. \quad (5.18)$$

Then, the equations of motion are given by

$$\rho \frac{\partial^2 s_x}{\partial t^2} = \frac{\partial T_{xx}}{\partial x} + \frac{\partial T_{xz}}{\partial z} \quad (5.19)$$

$$\rho \frac{\partial^2 s_y}{\partial t^2} = \frac{\partial T_{yx}}{\partial x} + \frac{\partial T_{yz}}{\partial z} \quad (5.20)$$

$$\rho \frac{\partial^2 s_z}{\partial t^2} = \frac{\partial T_{zx}}{\partial x} + \frac{\partial T_{zz}}{\partial z}, \quad (5.21)$$

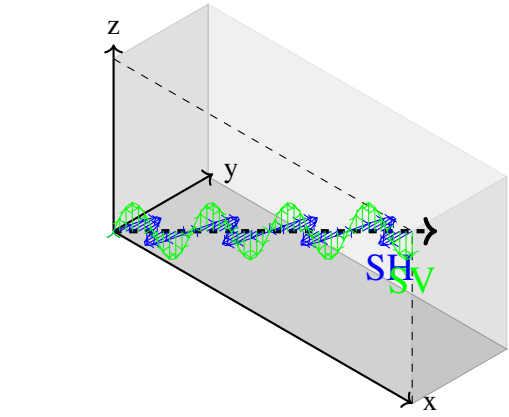


Figure 5.6: Propagations of SH wave and SV waves and the polarization.

and Hooke's law is given by

$$T_{xx} = (\lambda + 2\mu) \frac{\partial s_x}{\partial x} + \lambda \frac{\partial s_y}{\partial y} + \lambda \frac{\partial s_z}{\partial z} \quad (5.22)$$

$$T_{xz} = \mu \left(\frac{\partial s_x}{\partial z} + \frac{\partial s_z}{\partial x} \right) \quad (5.23)$$

$$T_{yx} = \mu \left(\frac{\partial s_y}{\partial x} + \frac{\partial s_x}{\partial y} \right) \quad (5.24)$$

$$T_{yz} = \mu \left(\frac{\partial s_y}{\partial z} + \frac{\partial s_z}{\partial y} \right) \quad (5.25)$$

$$T_{zz} = \lambda \frac{\partial s_x}{\partial x} + \lambda \frac{\partial s_y}{\partial y} + (\lambda + 2\mu) \frac{\partial s_z}{\partial z}. \quad (5.26)$$

Let us drop the stress term and rearrange the equations as,

$$\rho \partial_t^2 s_x = (\lambda + \mu) \partial_x (\partial_x s_x + \partial_z s_z) + \mu (\partial_x^2 s_x + \partial_z^2 s_x) \quad (5.27)$$

$$\rho \partial_t^2 s_y = +\mu (\partial_x^2 s_y + \partial_z^2 s_y) \quad (5.28)$$

$$\rho \partial_t^2 s_z = (\lambda + \mu) \partial_z (\partial_x s_x + \partial_z s_z) + \mu (\partial_x^2 s_z + \partial_z^2 s_z) \quad (5.29)$$

s_x and s_y are coupled with each other, whereas, s_y is decoupled with the others. Because s_y is polarized in a horizontal plane, the wave is called an SH wave. On the other hand, because s_x and s_z are composed of the P wave and vertical polarized S wave. the wave is called as P-SV wave. As described in later sections, the separation of the SH wave field and P-SV wave field is possible for a stratified Earth mode. Therefore, when we analyze seismic waveforms, rotations of horizontal components from north-south component and east-west component to transverse component (perpendicular to the great circle path between the source and the receiver) and radial component (parallel to the path). The transverse component represents the SH wave, and the radial component represents P-SV wave^{note 10)}.

5.3.2 Plane waves in the case of P-SV and SH waves: how to take the vector potential

When considering seismic wavefields, in particular, plane waves in the P-SV case (see next section for details), it is useful to introduce a potential as explained in the 3.4.1 section. As already mentioned, there is one degree of freedom in the vector potential. When considering wave propagation in a horizontal multilayer structure in Cartesian coordinates

$$\underset{SH}{A} = \begin{pmatrix} 0 \\ 0 \\ \chi \end{pmatrix} + \nabla \times \underset{SV}{\begin{pmatrix} 0 \\ 0 \\ \psi \end{pmatrix}} \quad (5.30)$$

SV and SH can be separated,⁽⁶⁾ and the outlook improves if we use the following formula. Because of the arbitrary property of the vector potential, we can choose a convenient way to take the vector potential for our problem. For future calculations, we will write down the case of propagation along a two-dimensional xz plane. Assuming that the potential does not depend on y , the partial derivative with respect to y disappears, and we have a simple form. Below unexplained components are set to zero.

^{note 10)}Exactly speaking, this separation valid for the far field.

P wave

$$s_x = \frac{\partial \phi}{\partial x}, \quad s_y = 0, \quad s_z = \frac{\partial \phi}{\partial z} \quad (5.31)$$

$$E_{xx} = \frac{\partial^2 \phi}{\partial x^2}, \quad E_{xz} = \frac{\partial^2 \phi}{\partial x \partial z}, \quad E_{zz} = \frac{\partial^2 \phi}{\partial z^2} \quad (5.32)$$

$$T_{xx} = E \frac{(1-\nu) \frac{\partial^2 \phi}{\partial x^2} + \nu \frac{\partial^2 \phi}{\partial z^2}}{(1+\nu)(1-2\nu)}, \quad T_{xz} = \frac{E}{1+\nu} \frac{\partial^2 \phi}{\partial x \partial z}, \quad T_{zz} = E \frac{(1-\nu) \frac{\partial^2 \phi}{\partial z^2} + \nu \frac{\partial^2 \phi}{\partial x^2}}{(1+\nu)(1-2\nu)} \quad (5.33)$$

SV wave

$$s_x = \frac{\partial^2 \psi}{\partial x \partial z}, \quad s_y = 0, \quad s_z = \frac{\partial^2 \psi}{\partial x^2} \quad (5.34)$$

$$E_{xx} = \frac{\partial^3 \psi}{\partial x^2 \partial z}, \quad E_{xz} = \frac{1}{2} \left(\frac{\partial^3 \psi}{\partial x \partial z^2} - \frac{\partial^3 \psi}{\partial x^3} \right), \quad E_{zz} = -E_{xx} \quad (5.35)$$

$$T_{xx} = -E \frac{(2\nu-1) \frac{\partial^3 \psi}{\partial x^2 \partial z}}{(1+\nu)(1-2\nu)}, \quad T_{xz} = \frac{E}{1+\nu} \frac{1}{2} \left(\frac{\partial^3 \psi}{\partial x \partial z^2} - \frac{\partial^3 \psi}{\partial x^3} \right), \quad T_{zz} = -T_{xx} \quad (5.36)$$

SH wave

$$s_x = 0, \quad s_y = -\frac{\partial \chi}{\partial x}, \quad s_z = 0 \quad (5.37)$$

$$E_{xx} = 0, \quad E_{xy} = -\frac{1}{2} \frac{\partial^2 \chi}{\partial x^2}, \quad E_{yz} = -\frac{1}{2} \frac{\partial^2 \chi}{\partial x \partial z}, \quad (5.38)$$

$$T_{xx} = 0, \quad T_{xz} = -\frac{E}{1+\nu} \frac{1}{2} \frac{\partial^2 \chi}{\partial x^2}, \quad T_{yz} = -\frac{E}{1+\nu} \frac{1}{2} \frac{\partial^2 \chi}{\partial x \partial z} \quad (5.39)$$

In the following, the plane wave amplitudes are treated as variables (e.g., A , B , C). In the next section, we will consider the potentials

$$\phi = \frac{A}{\omega i} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (5.40)$$

$$\psi = -\frac{B}{p_x \omega^2} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (5.41)$$

$$\chi = \frac{C}{\omega i} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (5.42)$$

and discuss the corresponding amplitudes.

§5.4 Reflection of SH-wave at a free surface

Let us consider the reflection and refraction of an SH wave in a semi-infinite elastic medium $z \leq 0$ with a free surface on $z = 0$. An incident SH wave entering with incident angle φ can be represented by a general solution of equation 5.28 as

$$s_y = Ae^{-i\omega(t-p_x x - p_z z)} + Be^{-i\omega(t-p_x x + p_z z)}, \quad (5.43)$$

where A and B are integral constants. The first term shows incident waves, and the second term shows the reflected wave. The free surface condition is written by

$$T_{zy} = -\mu \frac{\partial s_y}{\partial z} = 0. \quad (5.44)$$

Then the result $A = B$ indicates that the phase of the reflected wave is the same as that of the incident wave. Displacement on the free surface is given by,

$$s_y = 2Ae^{-i\omega(t-p_x x)}, \quad (5.45)$$

which is twice as large as the incident wave.

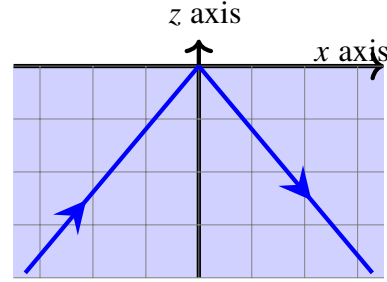


Figure 5.7:

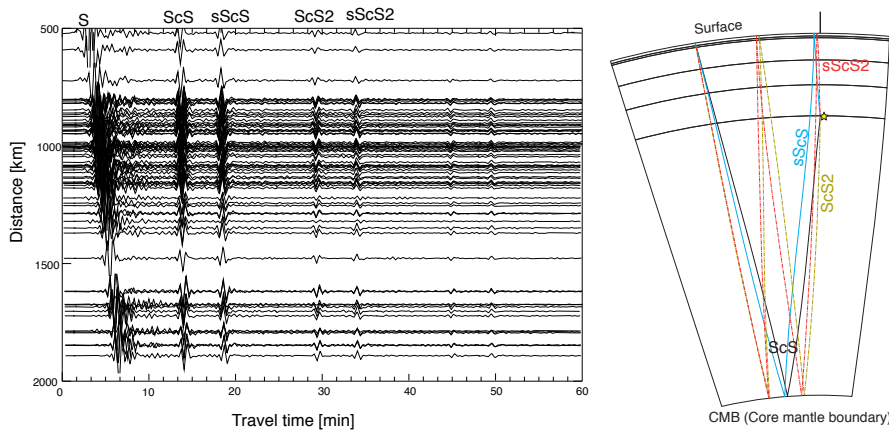


Figure 5.8: Left: An example of seismograms of ScS reverberations from the deep Ogasawara earthquake on 2015/5/30. They are vertical components of broadband seismometers of F-net band-pass-filtered from 30 s to 200 s. Right: Ray paths of ScS, sScS, ScS2, sScS2.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

This reflected wave can be interpreted as a "mirror image" of the incident wave to satisfy the boundary condition. When we consider an incident wave $Ae^{-i\omega(t-p_x x - p_z z)}$, superposition between a mirror described by $Ae^{-i\omega(t-p_x x + p_z z)}$ and the incident wave satisfy the boundary condition.

The governing equation for an SH wave is the same as the acoustic wave equation. However, their boundary conditions are different. For the SH wave, stress components T_{zx} and T_{zy} act as vector components, whereas the pressure for an acoustic wave is a scalar. To meet the free surface condition of an acoustic wave at $z = 0$, a mirror of $-Ae^{-i\omega(t-p_x x + p_z z)}$ is needed. The sign is different from the SH wave.

Figure 5.8 shows an example of reverberations of SH waves between the surface and the core-mantle boundary (CMB). Because the outer core is liquid, the CMB acts as a free surface for S waves. The figure shows many wave packets with fast apparent velocities, which correspond to multiple reflections of vertically propagating SH waves between the surface and the CMB^{note 11)}. The figure also shows that the phase of ScS (reflected once at the CMB) is the same as that of ScS2 (reflected twice).

^{note 11)} The amplitudes attenuate with distance due to intrinsic attenuation of the material.

§5.5 Reflection and conversion of P-SV wave at a free surface

For a given angular frequency ω , let us consider an incident P wave ($\mathbf{p}_\alpha^{in}$), and an incident SV wave ($\mathbf{p}_\beta^{in}$). The reflected P wave ($\mathbf{p}_\alpha^{refl}$), and the reflected SV wave ($\mathbf{p}_\beta^{refl}$) can be related by,

$$\mathbf{s} = \mathbf{A}_\alpha^{in} e^{i\omega(\mathbf{p}_\alpha^{in} \cdot \mathbf{x} - t)} + \mathbf{A}_\alpha^{refl} e^{i\omega(\mathbf{p}_\alpha^{refl} \cdot \mathbf{x} - t)} + \mathbf{A}_\beta^{in} e^{i\omega(\mathbf{p}_\beta^{in} \cdot \mathbf{x} - t)} + \mathbf{A}_\beta^{refl} e^{i\omega(\mathbf{p}_\beta^{refl} \cdot \mathbf{x} - t)}, \quad (5.46)$$

where $\mathbf{s}$ is displacement, and $\mathbf{A}$ represents polarization vector with the amplitudes. For simplicity, we solve a problem in the xz plane for propagating wave into y direction. The slowness vectors and the polarization vectors are given by,

$$\mathbf{p}_\alpha^{in} = \begin{pmatrix} p \\ \xi \end{pmatrix}, \quad \mathbf{p}_\alpha^{refl} = \begin{pmatrix} p \\ -\xi \end{pmatrix}, \quad \mathbf{p}_\beta^{in} = \begin{pmatrix} p \\ \eta \end{pmatrix}, \quad \mathbf{p}_\beta^{refl} = \begin{pmatrix} p \\ -\eta \end{pmatrix}, \quad (5.47)$$

$$\mathbf{A}_\alpha^{in} = A \begin{pmatrix} p \\ \xi \end{pmatrix}, \quad \mathbf{A}_\alpha^{refl} = B \begin{pmatrix} p \\ -\xi \end{pmatrix}, \quad \mathbf{A}_\beta^{in} = C \begin{pmatrix} \eta \\ -p \end{pmatrix}, \quad \mathbf{A}_\beta^{refl} = D \begin{pmatrix} \eta \\ p \end{pmatrix}, \quad (5.48)$$

Then we can simplify the equations as,

$$s_x(x, z) e^{i\omega(t - px)} = p(Ae^{i\omega\xi z} + Be^{-i\omega\xi z}) + \eta(Ce^{i\omega\eta z} + De^{-i\omega\eta z}) \quad (5.49)$$

$$s_z(x, z) e^{i\omega(t - px)} = \xi(Ae^{i\omega\xi z} - Be^{-i\omega\xi z}) + p(-Ce^{i\omega\eta z} + De^{-i\omega\eta z}). \quad (5.50)$$

Later in this chapter, we will explain the reflection and conversion coefficients of two cases: (1) incident SV wave ($A = 0$), and (2) incident P wave ($C = 0$) with boundary conditions of $T_{zx}(0) = 0$, and $T_{zz}(0) = 0$ as

$$2p\xi(A - B) + (\eta^2 + p^2)(C + D) = 0, \quad (5.51)$$

$$(\eta^2 - p^2)(A + B) - 2p\eta(C + D) = 0. \quad (5.52)$$

Here we define polarization vectors as

$$\hat{\mathbf{n}}_\alpha^{in} = \begin{pmatrix} p \\ \xi \end{pmatrix} \alpha, \quad \hat{\mathbf{n}}_\alpha^{refl} = \begin{pmatrix} p \\ -\xi \end{pmatrix} \alpha, \quad (5.53)$$

$$\hat{\mathbf{n}}_\beta^{in} = \begin{pmatrix} \eta \\ -p \end{pmatrix} \beta, \quad \hat{\mathbf{n}}_\beta^{refl} = \begin{pmatrix} \eta \\ p \end{pmatrix} \beta \quad (5.54)$$

In order to estimate the coefficients, we calculate the inner product between the polarization vectors and displacement $\mathbf{s}$.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

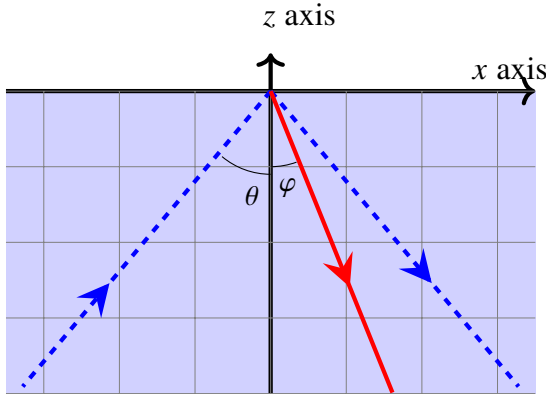


Figure 5.9: P-wave incidence

5.5.1 P-wave incidence

The reflection coefficient from P wave to P wave $R_{PP} = (B/\alpha)/(A/\alpha)$ is given by,

$$R_{PP} = -\frac{(\eta^2 - p^2)^2 - 4p^2\xi\eta}{(\eta^2 - p^2)^2 + 4p^2\xi\eta}. \quad (5.55)$$

The reflection coefficient from P wave to S wave $R_{PS} = (D/\beta)/(A/\alpha)$ is given by

$$R_{PS} = \frac{4p\xi(\eta^2 - p^2)}{(\eta^2 - p^2)^2 + 4p^2\xi\eta} \frac{\alpha}{\beta}. \quad (5.56)$$

Snell's Law

As already explained, both incident and reflected waves must have a dependence of $e^{-i\omega p x}$ to satisfy the boundary condition. ^{note 12)} In order for p to be a conserved quantity, the incident and reflected angles must satisfy the following relation: i.e., Snell's law,

$$\frac{\sin \varphi}{\beta} = \frac{\sin \theta}{\alpha}. \quad (5.57)$$

Conservation of energy

Consider the energy balance in a region on a thin region that includes the ground surface. The vertical energy flux must be balanced between the incident and reflected waves. The energy conservation law is given by

$$\alpha \cos \theta = \beta \cos \varphi |R_{PS}|^2 + \alpha \cos \theta |R_{PP}|^2. \quad (5.58)$$

^{note 12)} As explained in the ray theory chapter (??), p is a conserved quantity related to momentum.

Problem 5.2

1. Derive equation 5.55.
2. Derive equation 5.56.

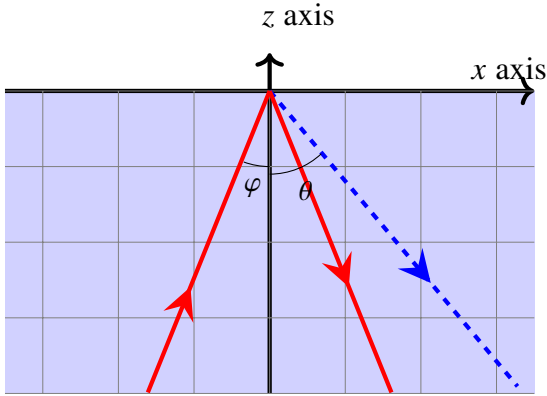
5.5.2 SV-wave incidence

Figure 5.10: SV-wave incidence

Consider the case of an SV wave with unit amplitude ($C = 1$) incident at an angle of 25° (φ in Figure 5.10). For this, both the reflected SV wave and the reflected P wave take non-zero values. To satisfy the boundary condition, the total stress must be zero after superposition. In fact, the resultant wavefield in the figure shows that the stress T_{zz} is zero at the surface. We search the values of B and D to meet the boundary condition: $T_{zz} = T_{xz} = 0$. Since the sum of the unknowns and the number of boundary conditions are equal, we can find B and D .

Reflection coefficient from S wave to S wave $R_{SS} = (D/\beta)/(C/\beta)$ is written by

$$R_{SS} = \frac{(\eta^2 - p^2)^2 - 4p^2\xi\eta}{(\eta^2 - p^2)^2 + 4p^2\xi\eta}. \quad (5.59)$$

Here slowness and the incidental angles are related as, $p = \frac{\sin \theta}{\alpha} = \frac{\sin \varphi}{\beta}$, $\xi = \frac{\cos \theta}{\alpha}$, $\eta = \frac{\cos \varphi}{\beta}$.

Reflection coefficient from S wave to P wave $R_{SP} = (B/\alpha)/(C/\beta)$ is given by,

$$R_{SP} = \frac{4p\eta(\eta^2 - p^2)}{(\eta^2 - p^2)^2 + 4p^2\xi\eta} \frac{\beta}{\alpha}. \quad (5.60)$$

Corresponding conservation of energy is represented by

$$\beta \cos \varphi = \beta \cos \varphi |R_{SS}|^2 + \alpha \cos \theta |R_{SP}|^2. \quad (5.61)$$

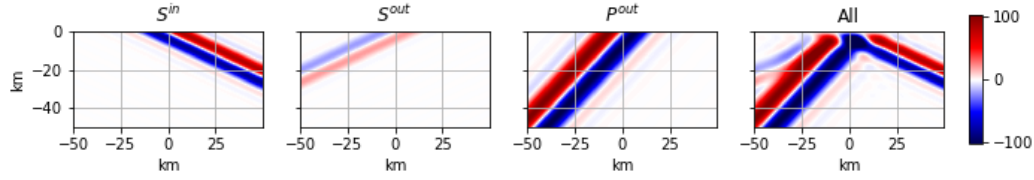


Figure 5.11: The stresses T_{zz} for reflected SV and reflected P waves when the SV wave is incident at an angle of incidence (φ) of 25° . This figure shows that the stress for the superposed wave field is zero at the ground surface.

Figure 5.12 shows the reflection coefficients.

Snell's law

Similarly, the continuity of displacement and stress at the boundary leads to Snell's law. The conservation of p and Snell's law hold as well as P-wave:

$$\frac{\sin \varphi}{\beta} = \frac{\sin \theta}{\alpha}. \quad (5.62)$$

Critical angle

After the critical angle φ_c

$$\varphi_c = \sin^{-1} \frac{\beta}{\alpha}, \quad (5.63)$$

total reflection occurs. In this case, the z component of the slowness of the converted P wave becomes imaginary and cannot carry net energy in the vertical direction (known as an inhomogeneous wave). The S-wave will be out of phase due to the inhomogeneous P-wave sticking near the ground surface. ^{note 13)}

Inhomogeneous P wave

Let us consider the stress T_{zz} for the case where the SV wave is incident at an angle of 75° (φ) beyond the critical angle. Figure 5.13 shows the incident SV wave and the reflected SV wave. The figure also shows that the phase of the reflected SV wave is out of phase by about 90 degrees (the incident SV wave is shaped like a half period of the sin function, while the reflected SV wave is shaped like one period of the cos function). The figure also shows that there are inhomogeneous P waves to meet the boundary conditions because of the phase shift of the reflected SV. The

^{note 13)}As calculated in Problem 5.1, the inhomogeneous waves carry energy up and down at a local scale but zero on a larger scale. When the wave incidents after the critical angle, the inhomogeneous P-wave accompanies the S-wave at the surface. The inhomogeneous P wave receives energy vertically from the S-wave and returns the energy with a slight time delay. The S wave is out of phase because of the temporary energy transfer by the inhomogeneous P wave

inhomogeneous P-wave is localized at the reflection point, and we can see that it is consistent with $T_{zz} = 0$.

Conservation of energy

Let us consider the energy balance in a thin region along the Earth's surface. The vertical energy flux must be balanced between the incident and reflected energy flux. Before the critical angle, the energy conservation law is given by

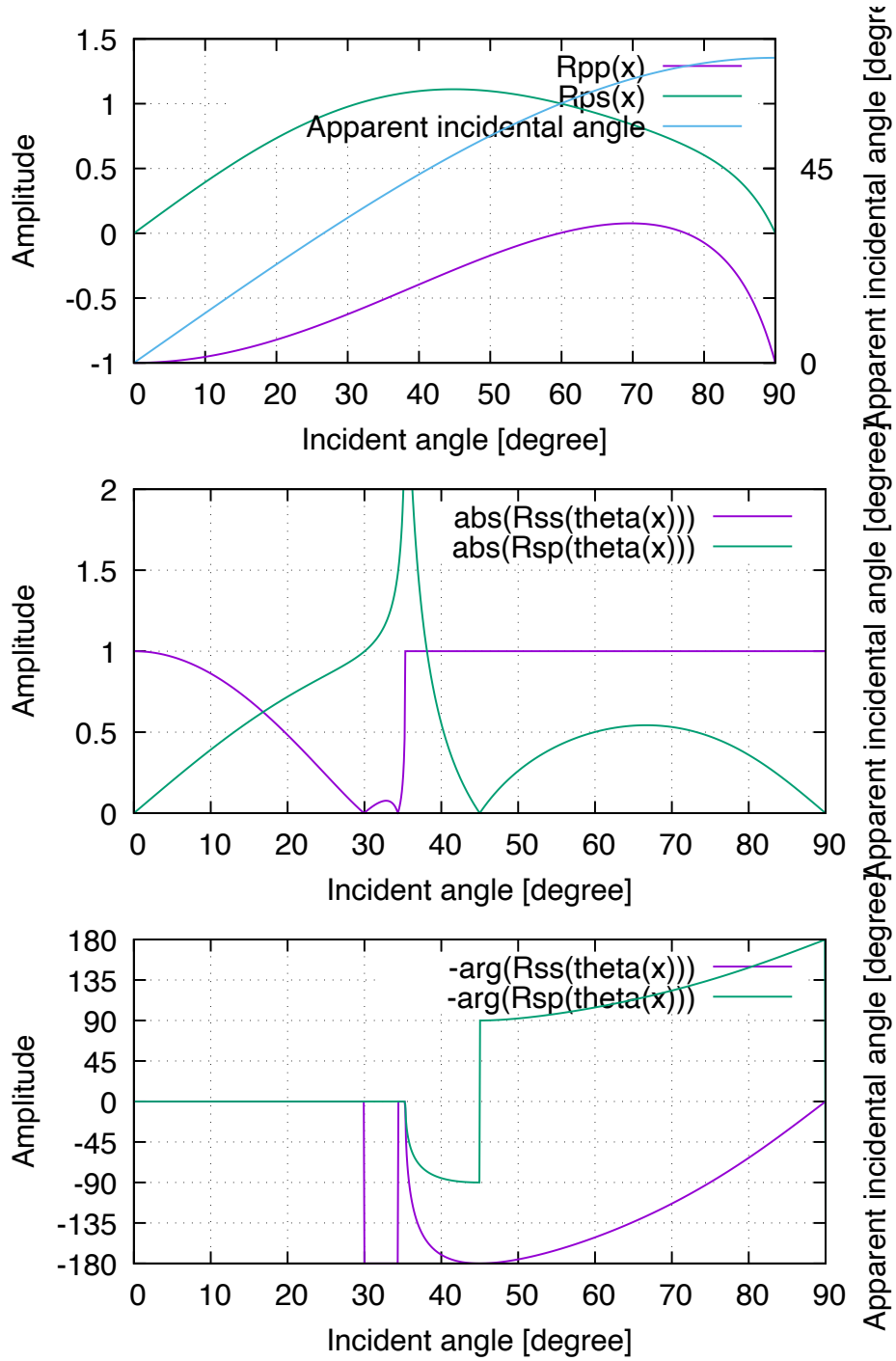
$$\beta \cos \varphi = \beta \cos \varphi |R_{SS}|^2 + \alpha \cos \theta |R_{SP}|^2. \quad (5.64)$$

Since the reflected wave becomes evanescent after the critical angle, the net energy flux becomes 0, which leads to the following relation,

$$1 = |R_{SS}|^2. \quad (5.65)$$

Problem 5.3

1. Derive equation 5.59.
2. Derive equation 5.60.

Figure 5.12: R_{sp} has the finite value at its critical angle.

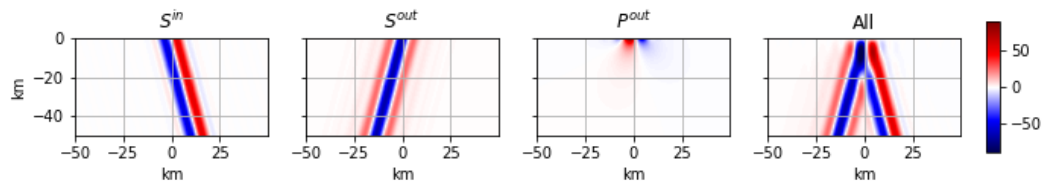


Figure 5.13: Stress T_{zz} for SV wave incident with angle φ of 75° , the reflected SV wave and the reflected P wave. wave. The superimposed stress field satisfies the boundary condition of the free surface.

Similarities and differences between SH and P waves

Both SH and P waves can be described by a single variable and behave very similarly because they satisfy the scalar wave equation. However, the reflection coefficient changes because the boundary conditions appear differently. Let us focus on this point here.

Here we consider an acoustic and an SH wave propagated in a xz plane. The governing equations are given by

$$\rho \frac{\partial^2 p}{\partial t^2} = \kappa \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} \right) \quad (5.66)$$

$$\rho \frac{\partial^2 s_y}{\partial t^2} = \mu \left(\frac{\partial^2 s_y}{\partial x^2} + \frac{\partial^2 s_y}{\partial y^2} \right). \quad (5.67)$$

The equations show that both can be described by a two-dimensional scalar wave equation. For example, if the speed of sound and SH wave velocity is the same, p and s_y have the same solution.

One point should be noted, however. The acoustic wave takes pressure as a variable, while the SH wave takes displacement as a variable. There is a big difference when considering free surface. When considering acoustic waves, $p = 0$ is the boundary condition at the free surface. On the other hand, when we consider a free surface at $z = 0$, the boundary condition at the free surface for SH waves is given by

$$-\rho \frac{\partial s_y}{\partial z} = 0. \quad (5.68)$$

This corresponds to the boundary condition of a rigid wall for acoustic waves. The reflection coefficient at the surface with respect to pressure for acoustic waves is -1, while for SH waves (with respect to displacement), the reflection coefficient at the surface is 1. The difference is whether stress or displacement is taken as the variable; in both cases, the stress at the surface is zero. The analogy with acoustic waves is valid, but be careful how you choose the variables.

Problem 5.4

Let us consider P wave propagation in a 3-D half space of fluid.

1. As in the case of SH wave, estimate reflection coefficients for an incident plane wave (P-wave) as a function of incident angle.
2. Compare the above result with P-wave reflection coefficients for the P-SV problem. In particular, discuss it for the incidental angles of about 0° and 90° .
3. As in the case of full space (see subsection 3.4.2), estimate the Green's function for an explosion source in a half-space of fluid. Discuss the behavior as the source depth approach the free surface.

5.5.3 Apparent incident angle of a P-SV wave

When analyzing seismic waveforms, particle motion is informative. For example, Figure 5.5.3 shows the particle motion of P wave against radial and vertical components when the deep earthquake at Ogasawara was recorded by the F-net station at Fukue. The figure shows an inclined linear polarization. The inclination shows the approximate incident angle. This subsection describes this relation.

First, let us consider that the P wave enters the free surface. The ratio between the vertical displacement s_x and the horizontal one s_y is given by

$$\frac{s_x}{s_z} = \frac{p(A+B) + \eta D}{\xi(A-B) + pD} = \frac{2p\eta}{\eta^2 - p^2} = \tan 2\varphi. \quad (5.69)$$

The inclination of P wave polarization (or apparent incident angle) θ' is twice as large as the S-wave reflected angle as (Figure 5.12),

$$\theta' = 2\varphi. \quad (5.70)$$

When the incident angle is sufficiently small for a Poisson medium ($\alpha = \sqrt{3}\beta$), the relation between the apparent incident angle and the P-wave incident angle can be simplified as

$$\theta' = 2\varphi \sim \frac{2\beta}{\alpha}\theta \sim 1.15\theta. \quad (5.71)$$

This result shows that the P-wave incident angle can be approximated by the apparent incident angle.

In the same manner, the incident angle of the S wave can be related to the apparent incident angle as:

$$\frac{s_x}{s_z} = \frac{pB + \eta(C+D)}{-\xi B + p(-C+D)} = -\frac{\eta^2 - p^2}{2p\xi} \quad (5.72)$$

When the incident angle is small (this assumption is valid for teleseismic events), the apparent incident angle can be related to the incident angle by

$$\varphi' = 2\frac{\beta^2}{\alpha^2}\varphi = 2\frac{\beta}{\alpha}\varphi. \quad (5.73)$$

Here we assumed that the incident angle is smaller than the critical angle.

It may seem somewhat intuitive that the direction of oscillation of the P wave coincides with the direction of incidence, but it is by no means obvious. For example, consider the limit $\beta \rightarrow 0$, assuming a medium similar to a fluid. In this case, the direction of oscillation is 0 degrees. Let us consider the case of a fluid. In Fig 5.15, we consider a pressure source in the ground. To satisfy

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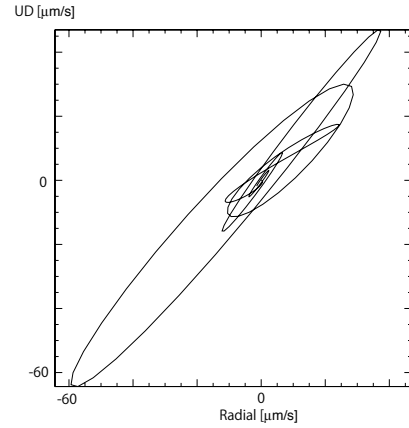


Figure 5.14: An example of particle motion of P-wave against radial and vertical components of F-net station at Fukue.

the boundary conditions at the water surface, we consider a pressure source with the opposite sign at the mirror-symmetric location. In this case, considering the particle trajectory at the ground surface, it will move up and down. Since this relationship holds at any time, the particle trajectory at the water surface is always vertical, regardless of the angle of incidence.

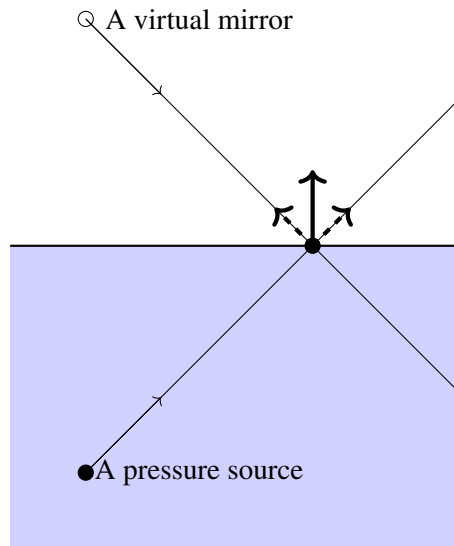


Figure 5.15: Particle motion near the surface in fluid.

Problem 5.5

1. Derive equation 5.69.
2. Derive equation 5.72.

§5.6 Rayleigh wave

In the previous sections, this chapter explained body waves that can propagate freely in the vertical direction. Inhomogeneous waves accompany incoming SV waves beyond the critical angle. Can inhomogeneous waves exist on their own without body waves? Rayleigh waves have energy concentrated near the surface and are known to propagate horizontally. Figure 5.16 shows the observation record during the 2014 Chilean earthquake. The waves labeled R1 and R2 are Rayleigh waves, which propagate at an approximately constant speed. The Chilean earthquake excited Rayleigh waves efficiently because the depth of the epicenter is 35 km, which is much shorter than the wavelength of the waves. First, let us consider qualitatively the nature of Rayleigh waves.

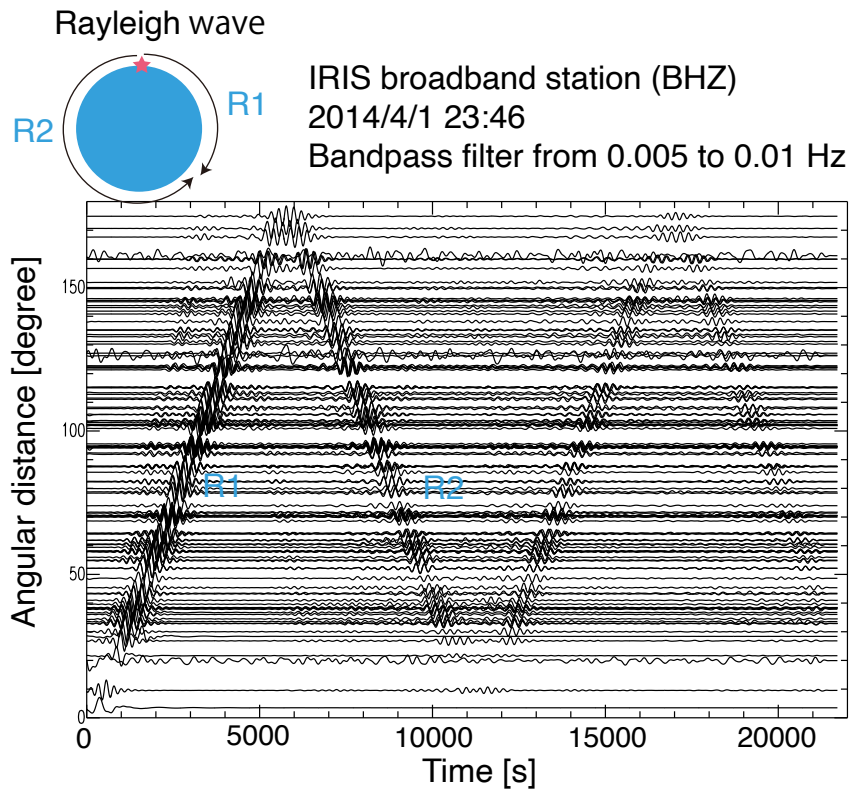


Figure 5.16: Waveforms in vertical components recorded by broadband seismometers when the 2014 Chilean earthquake.

When considering acoustic or SH wave propagation in a half-space, inhomogeneous waves cannot exist alone because they cannot satisfy the boundary conditions at the free surface. On the other hand, they can exist for P-SV propagations: these are known as Rayleigh waves.⁽⁵⁾ Rayleigh waves satisfy the free surface condition because P-wave-like and S-wave-like deformations can exist simultaneously, balancing each other at the surface. Rayleigh waves have energy concentrated near the surface so that we can observe global propagation (Fig. 5.17). We will first consider its properties qualitatively. First, let us consider the Rayleigh wave ^{note 14)}.

Let us consider a situation where an S-wave propagates to the right while oscillating vertically through an infinite medium. We then cut out an infinite medium in the middle. Then, to satisfy the boundary condition of zero stress at the free surface, the regions with large stress will bulge out (the parts with large volumetric strain). The free surface causes a large deformation, and the effective elastic constant decreases. As a result, the Rayleigh wave propagates slower than the S-wave (about 90% of the S-wave velocity). The particle motion becomes elliptical because the phase of the volume deformation is shifted by 90 degrees compared to that of the S-wave. Because the horizontal propagation velocity is slower than the S-wave and P-wave velocities, they can only exist as inhomogeneous waves. The amplitude decreases exponentially in the depth direction. At the surface, the particle motion is in the opposite direction of the rotation of a bicycle wheel (retrograde). On the other hand, in deeper regions (blue in the figure), the direction is the same as the rotation of a bicycle wheel (prograde).

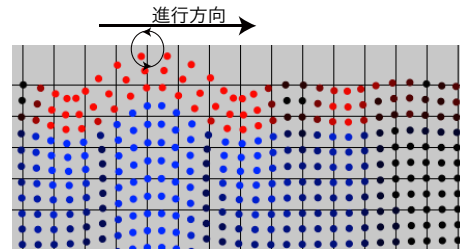


Figure 5.17: Schematic figure of Rayleigh wave propagation. Red circles show retrograde particle motions, whereas blue ones show prograde motions.

5.6.1 Can elastic waves along a free surface exist?

To consider this a little more quantitatively, let us now clarify the problem. Let us consider the propagation in the xz plane and assume that it does not change in the y direction, just as we would consider the reflection and conversion of a P-SV wave at the free surface. The boundary condition is given by $T_{zz} = T_{zx} = 0$ at the surface. To begin with, let us simply consider if acoustic waves and SH waves can propagate horizontally in half space.

As you may recall from Figure 5.15, the reflection coefficient of an acoustic wave at a free surface is -1 . Because the incident and reflected waves cancel each other out, the acoustic wave is not excited when an excitation source is near the free surface. Since the sign of the mirror image is reversed in Figure 5.15, it can be interpreted that in the case of a very shallow pressure source, the waves are not excited because the source pressure cancels out with the mirror image pressure, which has the opposite sign. The fact that acoustic waves cannot be excited by an explosive source near the water surface may, at first glance, seem inconsistent with physical intuition. However, it makes physical sense when one considers that the elastic energy is zero due to the free boundary

^{note 14)} see demo http://www.eri.u-tokyo.ac.jp/knishida/Seismology/Rayleigh_wave.html

surface (see also the section 4.6).

On the other hand, in the case of SH waves, the amplitude doubles at the surface and is amplified since the reflection coefficient is 1 in the case of SH waves. In other words, horizontally propagating waves do exist in the case of SH waves. This difference is due to the difference in boundary conditions. As mentioned before, the difference between displacement and stress comes into play because the pressure satisfies the scalar wave equation for acoustic waves, while s_y satisfies it in the SH case. If so, is this possible for SV waves? It is impossible because SV waves do not satisfy the boundary conditions when horizontally propagating.

5.6.2 A case of the reflection coefficient of zero

Let us now reconsider the number of boundary conditions. For the $P - SV$ problem, we need two boundary conditions, $T_{zz} = 0, T_{zx} = 0$. If we now consider arbitrary incident waves, we will need to add together two independent solutions to eliminate these two. This situation corresponds exactly to the reflection/transformation wave for the incident wave at the free surface in the previous section. The similarity means that a combination of the three waves is required. So let us look at the SV incidence in Figure 5.12. R_{SS} in the figure shows two points where the amplitude of the S reflection is zero. R_{SP} in the figure shows one point where the P reflection is zero. The zeros show the possibility of combining the two waves to satisfy the boundary condition if one looks for the appropriate frequency. Let us consider the possibility of this kind of solution.

In the following, let us consider T_{zx}^P, T_{zz}^P caused by an inhomogeneous P wave and T_{zx}^S, T_{zz}^S caused by an inhomogeneous SV wave. If we can find a phase velocity or slowness p that gives the same stress ratio T_{zx}/T_{zz} for both waves, the difference between the two normalized solutions will satisfy the boundary conditions. Now, let us look at the stress ratio for each wave.

Inhomogeneous P wave

Here, the scalar potential for an inhomogeneous P wave is defined by

$$\phi = e^{i\omega p x} e^{-\omega \xi z} e^{-i\omega t}. \quad (5.74)$$

P-wave velocity α satisfies the relation $\alpha^{-2} = p^2 - \xi^2$. The result in section ?? leads to the ratio between T_{zz} and T_{zx} :

$$\frac{T_{zx}}{T_{zz}} = -i \frac{(1-2\nu)\xi p}{(1-\nu)\xi^2 - \nu p^2}. \quad (5.75)$$

For a Poisson solid ($\nu = 1/4$), we can simplify the relations by eliminating ξ :

$$\frac{T_{zx}}{T_{zz}} = -i \frac{2\sqrt{p^2 - \alpha^{-2}} p}{2p^2 - 3\alpha^{-2}}. \quad (5.76)$$

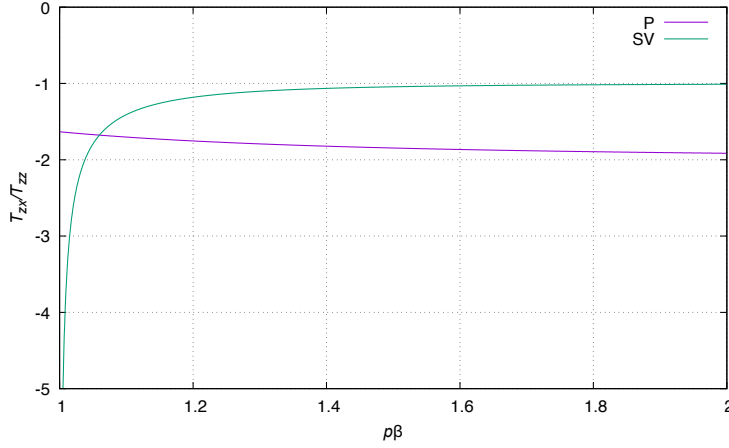


Figure 5.18: Stress ratio T_{zx}/T_{zz} at the surface for slowness p . Here we consider a Poisson material, and the slowness is normalized by the S-wave velocity. We plot them for $p\beta > 1$ because the vertical wavenumber becomes pure imaginary (which can propagate in the vertical direction as body waves) for $p\beta < 1$.

Inhomogeneous SV wave

Here, the component of vector potential for an inhomogeneous SV wave is defined by

$$\frac{\partial \psi}{\partial x} = e^{i\omega p x} e^{-\omega \eta z} e^{-i\omega t}. \quad (5.77)$$

The S-wave velocity β satisfies the relation $\beta^{-2} = p^2 - \eta^2$. The result in section ?? leads to the stress ratio between T_{zz} and T_{zx} as:

$$\frac{T_{zx}}{T_{zz}} = -i \frac{(\eta^2 - p^2)}{2p\eta}, \quad (5.78)$$

For a Poisson solid ($\alpha = \sqrt{3}\beta$), elimination of η leads to

$$\frac{T_{zx}}{T_{zz}} = -i \frac{2p^2 - \beta^{-2}}{2p\sqrt{p^2 - \beta^{-2}}}. \quad (5.79)$$

Let us compare the two stress ratios. Figure 5.18 shows that the two curves intersect when p is slightly greater than β^{-1} , which would represent the condition under which inhomogeneous waves can exist alone. This corresponds to Rayleigh waves. Let us consider in more detail the conditions under which the stress ratio at the surface is constant. The condition of equal stress ratio leads to the following relation;

$$(\eta^2 - p^2)^2 + 4p^2 \xi \eta = 0. \quad (5.80)$$

For the better understandings of the equation, we define $X \equiv p^2\beta^2$ and $\gamma^2 \equiv \alpha^{-2}/\beta^{-2}$. With the squared stress ratios of the inhomogeneous P wave and the inhomogeneous SV wave, we can rearrange the equation is as follows:

$$16(1 - \gamma^2)X^3 - (24 - 16\gamma^2)X^2 + 8X - 1 = 0. \quad (5.81)$$

In the case of Poisson material, the equation can be simplified as,

$$32X^3 - 56X^2 - 24X - 3 = 0, \quad (5.82)$$

and the analytic solutions are given by

$$X = \frac{1}{4}, \frac{3}{4} \pm \frac{\sqrt{3}}{4}. \quad (5.83)$$

Only the last one satisfies the condition $p > \beta^{-1}$, and the solution corresponds to the solution shown by Figure 5.18. What are the physical meanings of the other two solutions? The incident angles estimated from p are 30° and 34.3° for Poisson material. These two angles correspond exactly to the zero crossing of the R_{SS} in Figure 5.12: i.e., the case where all incident S waves are converted to reflected P waves. Thus, the three solutions make physical sense. In other words, the problem of finding a condition where the incident S-wave reflected at the free surface is extended to the problem of finding a condition where the inhomogeneous P-wave converted only the inhomogeneous SV-wave wave. We summarize the displacement of the Rayleigh wave:

$$s_x(x, z, t) = ipB \left[e^{\omega\xi z} + \frac{1 - 2p\beta}{2p\beta} e^{\omega\eta z} \right] e^{i\omega(t - px)} \quad (5.84)$$

$$s_z(x, z, t) = \xi B \left[e^{\omega\xi z} + \frac{2p\beta}{1 - 2p\beta} e^{\omega\eta z} \right] e^{i\omega(t - px)}, \quad (5.85)$$

where $\xi = \sqrt{p^2 - \alpha^{-2}}$, $\eta = \sqrt{p^2 - \beta^{-2}}$, and B is an integral constant. Also, in order to choose a physically meaningful solution (no divergence at $z = -\infty$), the signs are chosen so that $\text{Im } \eta < 0$ and $\text{Im } \xi < 0$.

Figure 5.19(a) shows the depth profile of Rayleigh wave amplitudes (s_x and s_z). It can be seen that the amplitude decreases exponentially with depth. The major difference from a body wave is the phase shift between horizontal and vertical motion. The motion of particles at the surface is plotted in Fig 5.19(b). Taking the real parts of s_x and s_z , we see that they rotate in the direction of the arrows in the figure since $s_x \propto -\sin \omega t$, $s_z \propto \cos \omega t$. Since the Rayleigh wave is now traveling from left to right in the figure, we call it retrograde. ^{note 15} Also, the sign of s_x reverses around $z = -0.25$ (normalized depth), so the direction of rotation is reversed at deeper points. This direction is called the forward direction (prograde).

To better visualize this, a demo page for Rayleigh waves is available on the web. ^{note 16} Please refer to it as needed.

^{note 15} It is easy to understand if you think of the direction of rotation of a bicycle wheel.

^{note 16} https://www.eri.u-tokyo.ac.jp/people/knishida/Seismology/Rayleigh_wave.html

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

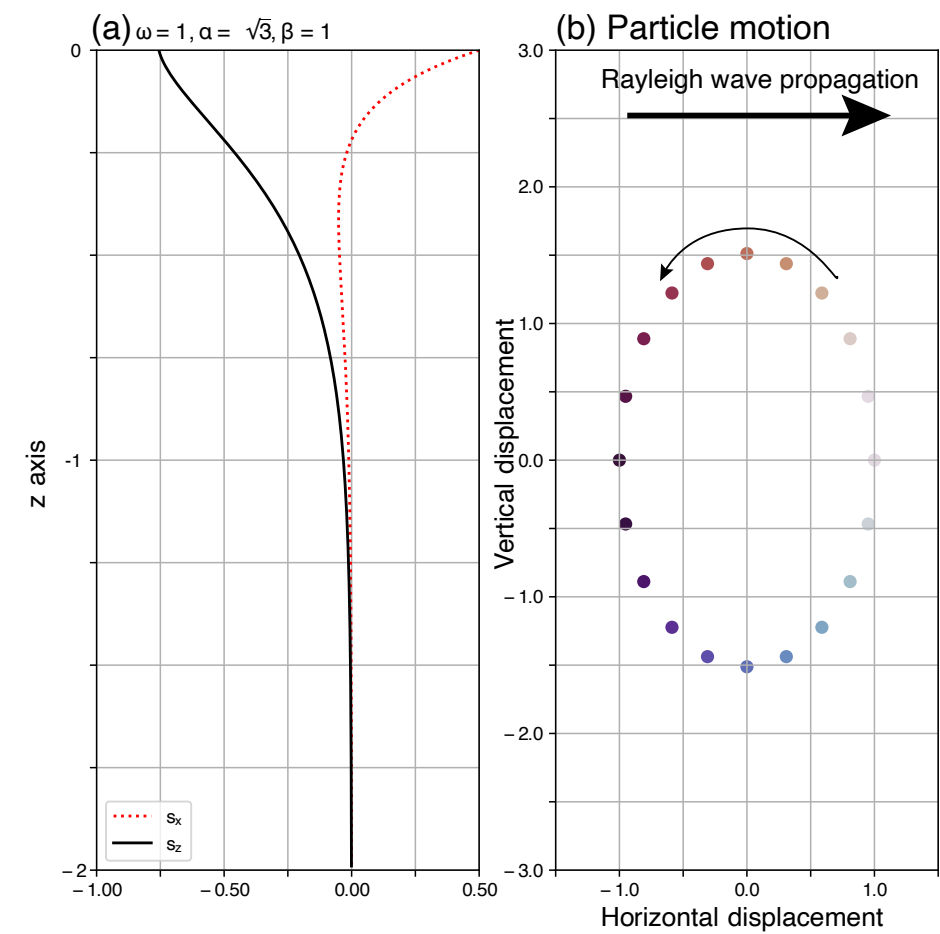
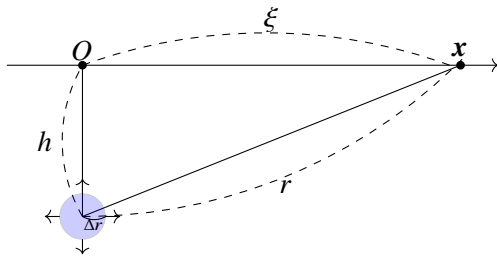


Figure 5.19: (a) Depth profile of Rayleigh wave amplitudes. (b) Particle motion at the surface.

Rayleigh waves are two-dimensional because they have energy only near the surface. The behavior is, therefore, similar to that of a two-dimensional Green's function. Specifically, the amplitude is proportional to $r^{-1/2}$. ^{note 17)} When the epicentral distance is far, the surface waves are larger than the body waves. ^{note 18)} Also, surface waves are not efficiently excited when the hypocenter is deeper than the wavelength (because they decay exponentially in the depth direction).

§5.7 Excitation of Rayleigh waves



Pressure source

Figure 5.20: Configuration and coordinate system of an explosive source (pressure source) in a semi-infinite medium.

When considering the excitation of Rayleigh waves, it is difficult to intuitively understand how they are excited. As long as we think in terms of the superposition of plane waves, considering reflected and converted waves does not seem to lead to the generation of Rayleigh waves just to satisfy boundary conditions. What becomes important here is the near-field term introduced in the section on Green's function for elastic bodies. We can physically explain this by considering that the near-field term creates a secondary excitation source to satisfy the boundary conditions, and this secondary excitation source excites the Rayleigh wave. Here, let us consider an explosive source as an example. In the following, we first separate the stress field generated by the explosive source into the near-field and far-field terms. Next, we reduce it to a problem of inhomogeneous wave incidence using Weyl's integral representation, and consider the Rayleigh wave as a problem of incidence and reflection of inhomogeneous waves.

^{note 17)} Roughly speaking, for epicentral distance r , amplitude A , and propagation velocity c_R , the energy flux can be written as $c_R \rho \omega A^2$, and energy conservation yields

$$\nabla \cdot \mathbf{K} = \frac{1}{r} \frac{\partial}{\partial r} (Kr) = 0, \quad (5.86)$$

which leads to $A \sim r^{-1/2}$.

^{note 18)} Recalling the far-field term of Green's function, we can see that the body waves are proportional to r^{-1} . This can be derived from considering the conservation of energy as well as surface waves.

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5.7.1 Wavefield and stress field produced by an explosive source

The frequency component of displacement $S(\omega)$ can be written as

$$S_r(\omega) = \frac{\Delta p \Delta r^3}{3(\lambda + 2\mu)} \frac{\omega}{\alpha^2} e^{i\omega \frac{r}{\alpha}} \left(\underbrace{-\frac{i}{(kr)^2}}_{\text{Near-field}} + \underbrace{\frac{1}{kr}}_{\text{Far-field}} \right), \quad (5.87)$$

The stresses T_{zz} and T_{zx} at the surface are respectively given by

$$T_{zz} = \frac{\Delta p \Delta r^3}{3(\lambda + 2\mu)} e^{i\omega \frac{r}{\alpha}} \left(\frac{2\mu(\xi^2 - 2h^2)}{r^5} \frac{1}{i\omega} - \frac{\lambda r^2 + 2\mu h^2}{\alpha r^4} \right) \quad (5.88)$$

and

$$T_{zx} = -\frac{\Delta p \Delta r^3}{3(\lambda + 2\mu)} e^{i\omega \frac{r}{\alpha}} \left(\frac{6\mu\xi h}{r^5 i\omega} - \frac{2\mu\xi h}{r^4} \right). \quad (5.89)$$

5.7.2 Contribution of the near-field term

For the far-field term, we have seen that reflected and converted waves are generated to satisfy boundary conditions. How does the near-field term behave? Here, let us look at the behavior of the near-field term of T_{zz} .

$$T_{zz} \propto \frac{e^{i\omega \frac{r}{\alpha}}}{i\omega} \left(\frac{\xi^2 - 2h^2}{r^5} \right) \quad (5.90)$$

In this case, the near-field term has sufficient amplitude. From the ξ dependence of T_{zz} , we can see that the secondary source is localized near the origin and has a short-wavelength spatial structure compared to the wavelength of interest. For such a short-wavelength structure to propagate horizontally, it must be an inhomogeneous wave that decays exponentially in the depth direction. This can be viewed as a problem where an inhomogeneous P-wave (or S-wave) is emitted horizontally from directly above the source and converted into an inhomogeneous S-wave (or P-wave) to satisfy the boundary conditions. Since $T_{zz} = 0$ must hold at the free boundary surface, we consider $-T_{zz}$, which cancels this out, as a secondary excitation source. Regarding this term, let us consider its contribution by comparing h with the wavelength.

$$h > \lambda$$

When h is sufficiently longer than the wavelength, this term can be ignored compared to the far-field term. In the first place, the near-field term plays a minor role in the wavefield.

$$h < \lambda$$

Let us consider the case where the depth of the explosive source is shallower than the wavelength. In this case, the near-field term has sufficient amplitude. From the ξ dependence of T_{zz} , we can see that it is localized near the origin. Therefore, when considering a certain frequency ω , the secondary source has a shorter wavelength spatial structure compared to the corresponding wavelength. To propagate horizontally, it must be an inhomogeneous wave that decays exponentially in the depth direction. Moreover, it can be viewed as a problem where an inhomogeneous P-wave (or S-wave) is excited horizontally from directly above the source and converted into an inhomogeneous S-wave (or P-wave) to satisfy the boundary conditions. When the source is shallow compared to the wavelength, both the near-field and far-field terms become important, and it becomes difficult to distinguish their contributions. Qualitatively, the contribution of the inhomogeneous wave becomes larger in the near-field term and plays an important role^{note 19)}. As can be seen from the fact that an inhomogeneous wave cannot propagate in the vertical direction, it is generally understood as a deformation in response to a forced variation. You can probably imagine that when it exceeds the critical angle, it cannot propagate vertically and is forced to vibrate to satisfy the boundary conditions. Rayleigh waves can also be interpreted as a pair of inhomogeneous waves that can exceptionally become **self-sustaining** without being forced. Therefore, let us consider it as an incidence problem of an inhomogeneous P-wave.

5.7.3 As a problem of inhomogeneous P-wave incidence

Here, let us interpret the Rayleigh wave as a problem of reflection and transmission of P-SV inhomogeneous waves. To consider the explosive source, let us consider the 3-D Green's function in a homogeneous infinite medium corresponding to the P-wave potential. The Green's function G^{3D} in the frequency domain satisfies

$$\frac{\omega^2}{\kappa} G^{3D} + \frac{1}{\rho_0} \nabla^2 G^{3D} = \delta(\mathbf{x}). \quad (5.91)$$

By performing a spatial Fourier transform and then an inverse Fourier transform again, we can see that it can be written as

$$G^{3D} = -\frac{1}{\kappa} \iiint \frac{\kappa e^{i(k_x x + k_y y + k_z z)}}{\alpha^2(k_x^2 + k_y^2 + k_z^2) - \omega^2} dk_x dk_y dk_z. \quad (5.92)$$

Here, we used the relation $\kappa = \rho_0 \alpha^2$. This equation indicates that the Green's function can be expressed by the superposition of plane waves.

If we assume that the z -component of the wavenumber is real, we cannot account for inhomogeneous waves. Therefore, let us consider an equation that does not Fourier-transform only in z :

$$\left(-\frac{1}{\rho_0} (k_x^2 + k_y^2) + \frac{\omega^2}{\kappa} \right) G^{3D} + \frac{1}{\rho_0} \frac{\partial^2 G^{3D}}{\partial z^2} = \delta(z). \quad (5.93)$$

^{note 19)}Strictly speaking, the far-field term also includes an inhomogeneous wave, but its effect is relatively small.

This can be written as above. Regard k_x, k_y as constants, and define k_z as

$$k_z \equiv \sqrt{\left(\frac{\omega}{\alpha}\right)^2 - k_x^2 - k_y^2}. \quad (5.94)$$

With this definition, it takes the same form as the Green's function for the 1-D wave equation with respect to z (equation (??)). We can write it as

$$G^{3D}(k_x, k_y, \omega) = \frac{\rho_0 i}{2k_z} e^{-ik_z|z|}. \quad (5.95)$$

Applying a 2-D inverse Fourier transform with respect to k_x and k_y , we get

$$G^{3D}(x, y, z, \omega) = \frac{\rho_0 i}{8\pi^2} \iint_{-\infty}^{\infty} dk_x dk_y \frac{e^{i(k_x x + k_y y - k_z |z|)}}{k_z}. \quad (5.96)$$

Comparing this with the 3-D Green's function for an infinite medium (equation (??)),

$$-\frac{\rho_0 i}{4\pi} \frac{e^{-ikr}}{r} = \frac{\rho_0}{8\pi^2} \iint_{-\infty}^{\infty} dk_x dk_y \frac{e^{i(k_x x + k_y y - k_z |z|)}}{k_z}. \quad (5.97)$$

This equation is also called Weyl's integral representation, which indicates that the 3-D infinite Green's function can be represented by the superposition of plane waves. Furthermore, although it implicitly assumes that k_x and k_y are real numbers, by separating z , it can be naturally extended to inhomogeneous waves in the z -axis direction. Therefore, by considering the reflection and transmission of plane waves at boundaries, it can naturally be extended to wave propagation in a two-layer medium.

Looking at Weyl's integral representation, you will see that it is represented by an inhomogeneous P-wave that attenuates upward from the source. Since the amplitude of this inhomogeneous P-wave diverges with depth, it seems physically impossible at first glance. Let us recall the representation theorem here. It can be realized by considering a boundary surface for the incident inhomogeneous P-wave and providing the boundary conditions of stress and displacement at the boundary surface^{note 20)}. In other words, we will consider it as a problem of incidence of an upwardly decaying inhomogeneous P-wave emitted from the source.

Let us take the horizontal slowness on the horizontal axis and calculate the reflection and conversion coefficients even for values larger than the S-wave slowness (Figure 5.21). When $p\beta$ is smaller than $1/\sqrt{3}$, the incident P-wave is a body wave, so it is equivalent to the problem of reflection and conversion of a P-wave at a free boundary surface. When $p\beta$ is greater than $1/\sqrt{3}$ and smaller than 1, the incident P-wave is an inhomogeneous wave, but the converted S-wave is a body wave. The inhomogeneous P-wave appearing as a reflection corresponds to a wave whose amplitude decreases vertically downward.

At the point where $p\beta$ is slightly larger than 1, R_{ps} and R_{pp} diverge^{note 21)}. You can see that the wavenumber corresponding to the Rayleigh wave is emphasized. What is happening?

^{note 20)} The representation theorem is powerful as it allows us to easily consider such virtual operations

^{note 21)} They diverge in the wavenumber domain, but for physically realizable excitations, they do not diverge in the spatiotemporal domain.

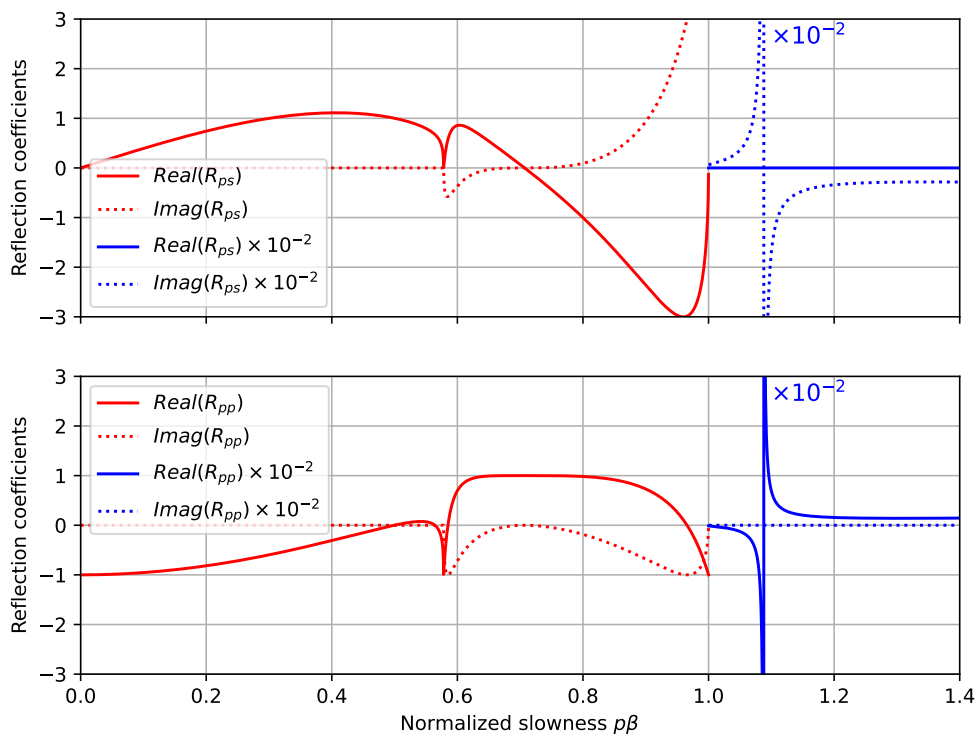


Figure 5.21: Reflection and transmission coefficients for inhomogeneous P-wave incidence. A wave decaying vertically upward is taken as the incident inhomogeneous P-wave. For plotting purposes, the reflection and conversion coefficients are multiplied by $1/100$ for $p\beta > 1$. It can be seen that they diverge at the velocity corresponding to the Rayleigh wave velocity.

Since it becomes difficult to grasp the physical meaning when it diverges, let us consider an inhomogeneous P-wave that decreases vertically downward^{note 22)}. Looking at Figure 5.22, you will notice that R_{pp} becomes 0 at the slowness corresponding to the Rayleigh wave velocity. Now, an inhomogeneous wave decreasing vertically upward can only exist if supported by an external force at the excitation source (which can also be rephrased as the near-field term), so if this component exists, it cannot exist far from the source. At the slowness corresponding to the Rayleigh wave velocity, it is possible to exist even without an excitation source because an inhomogeneous P-wave and an inhomogeneous S-wave decreasing vertically downward support each other. In this way, it can be viewed as a generalized problem of the reflection and conversion of inhomogeneous P-waves.

Let us consider what is happening physically. When the incident inhomogeneous P-wave reaches the surface, secondary waves are excited to satisfy the free boundary surface. From the representation theorem, consider placing a secondary pressure source that cancels out the stress of the incident inhomogeneous P-wave. To satisfy the boundary conditions, inhomogeneous P-waves and inhomogeneous S-waves are excited. Due to physical constraints, secondary waves can exist only as vertically downward decreasing waves. Therefore, except under the condition $R_{pp} = 0$, the incident inhomogeneous P-wave is required to support the secondarily excited inhomogeneous P-waves and S-waves. Thus, the near-field term is used to cause the surface to bulge, thereby satisfying the boundary conditions.

On the other hand, when the condition $R_{pp} = 0$ is satisfied, there is no need to support the reflected inhomogeneous P-wave and the converted inhomogeneous S-wave with the incident inhomogeneous P-wave. Therefore, a cycle is established where the inhomogeneous P-wave excites the inhomogeneous S-wave, and the inhomogeneous S-wave excites the inhomogeneous P-wave. Because it propagates maintaining its amplitude over long distances, it diverges at the wavenumber corresponding to the Rayleigh wave velocity, which has an amplitude when Fourier transformed^{note 23)}. In this way, the generation of Rayleigh waves can be understood by considering that the loop of inhomogeneous P-waves and inhomogeneous S-waves, whose phases are shifted by 90 degrees relative to each other, closes only at the Rayleigh wave velocity.

^{note 22)}When considering the incidence of an inhomogeneous P-wave descending vertically upward, we considered an inhomogeneous P-wave decreasing vertically downward as the reflected wave, and its conversion coefficient diverged at the Rayleigh wave velocity. Therefore, you can consider it as re-normalized by the amplitude of the converted inhomogeneous P-wave.

^{note 23)}Due to physical requirements, it will have a finite value if converted back to the time domain

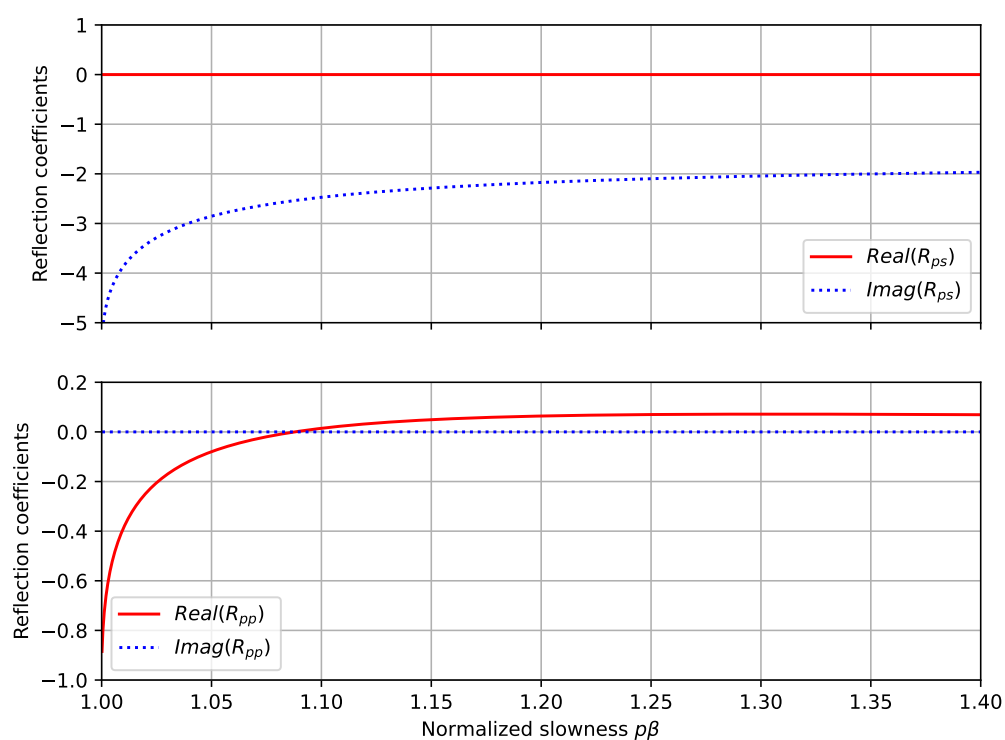


Figure 5.22: Reflection and transmission coefficients for inhomogeneous P-wave incidence. A wave decaying vertically downward is taken as the incident inhomogeneous P-wave. It can be seen that R_{pp} becomes zero at a point where the slowness is slightly greater than $1/\beta$.

Problem 5.6

1. Show that the equation 5.81 has a solution with slowness greater than $p = 1/\beta$ for any elastic constant.
2. Illustrate the Rayleigh wave velocity versus Poisson's ratio when $\beta = 1$. Also, discuss the physical meanings.
3. Calculate the ellipticity of the particle motion at the earth's surface in the same way and discuss it physically.

§5.8 Lamb's solution

We present the analytical form of Lamb's problem for a 3-D elastic half-space, as given by Kausel (2014).⁽²⁾ r is the radial distance, θ is the azimuth, μ is the shear modulus, ρ is the density, ν is Poisson's ratio, C_R is the Rayleigh wave velocity, C_S is the S-wave velocity, C_P is the P-wave velocity, t is time, a is C_S/C_P . Here, κ_j are the three dimensionless roots to the Rayleigh characteristic equation ($= C_S/C_j$), $\gamma \equiv \kappa_1 = C_S/C_R$ is the true Rayleigh root, and τ is the dimensionless time tC_S/r .

The Rayleigh characteristic equation is given by

$$16(1 - a^2)\kappa^6 - 8(3 - 2a^2)\kappa^4 + 8\kappa^2 - 1 = 0. \quad (5.98)$$

$$A_j = \frac{(\kappa_j^2 - \frac{1}{2})^2 \sqrt{a^2 - \kappa_j^2}}{D_j}, \quad B_j = \frac{(1 - 2\kappa_j^2)(1 - \kappa_j^2)}{D_j}, \quad C_j = \frac{(1 - \kappa_j^2)\sqrt{a^2 - \kappa_j^2}}{D_j} \quad (5.99)$$

$$D_j = (\kappa_j^2 - \kappa_i^2)(\kappa_j^2 - \kappa_k^2), \quad i \neq j \neq k$$

$$u_{zz}(r, \tau) = \frac{(1 - \nu)}{2\pi\mu r} \begin{cases} \frac{1}{2} \left(1 - \sum_{j=1}^3 \frac{A_j}{\sqrt{\tau^2 - \kappa_j^2}} \right), & a < \tau < 1 \\ 1 - \frac{A_1}{\sqrt{\tau^2 - \gamma^2}}, & 1 \leq \tau < \gamma \\ 1, & \tau \geq \gamma \end{cases} \quad (5.100)$$

$$u_{rx} = \frac{(\cos \theta)}{2\pi\mu r} \begin{cases} \frac{1}{2} (1 - \nu) \tau^2 \sum_{j=1}^3 \frac{C_j}{\sqrt{\tau^2 - \kappa_j^2}}, & a < \tau < 1 \\ 1 + (1 - \nu) \tau^2 \frac{C_1}{\sqrt{\tau^2 - r^2}}, & 1 \leq \tau < \gamma \\ 1, & \tau \geq \gamma \end{cases} \quad (5.101)$$

$$u_{\theta x} = \frac{(1 - \nu)(-\sin \theta)}{2\pi\mu r} \begin{cases} \frac{1}{2} \left[1 - \sum_{j=1}^3 C_j \sqrt{\tau^2 - \kappa_j^2} \right], & a < \tau < 1 \\ 1 - C_1 \sqrt{\tau^2 - \gamma^2}, & 1 \leq \tau < \gamma \\ 1, & \tau \geq \gamma \end{cases} \quad (5.102)$$

§5.9 Bibliography

- [1] F.A. Dahlen and J. Tromp. *Theoretical Global Seismology*. Princeton University Press, Princeton, 1998.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

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- [2] Eduardo Kausel. Lamb's problem at its simplest. *Proc. R. Soc. A Math. Phys. Eng. Sci.*, 469(2149):20120462–20120462, nov 2012.
- [3] K Nishida. Global propagation of body waves revealed by cross-correlation analysis of seismic hum. *Geophys. Res. Lett.*, 40(9):1691–1696, May 2013.
- [4] F. Omori. Note on the preliminary tremor of earthquake motion. *The journal of the College of Science, Imperial University of Tokyo, Japan*, 11(3):147–159, 1899.
- [5] John William Strutt Rayleigh. On waves propagated along the plane surface of an elastic solid. *Proc. Lond. Math. Soc.*, s1-17(1):4–11, November 1885.
- [6] Masanori Saitō. *The theory of seismic wave propagation*. 2016.